Object Oriented Programming (CSEG1044)

Unit 04 – Lecture 04: Swing Basics (Components, Containers, Layouts)

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Sources: Schildt, *Java: The Complete Reference* (12e) (generics, lambdas, Swing, JDBC).; Bloch, *Effective Java* (3e) (generics + lambdas).; Oracle Java Tutorials: Generics + Lambda Expressions + Swing + JDBC Basics (accessed February 27, 2026).; Java SE API docs (e.g., `java.util.function`, `javax.swing`, `java.sql`) (accessed February 27, 2026).

Quick Links

Overview Key Topics Demo Summary

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Agenda

1 Overview

2 Key Topics

3 Demo / Example

4 Summary

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Learning Outcomes

- Explain Swing component model: `JFrame`, `JPanel`, components vs containers.

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Learning Outcomes

- Explain Swing component model: `JFrame`, `JPanel`, components vs containers.
- Use layout managers (Flow/Border/Grid) to structure a UI.

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Learning Outcomes

- Explain Swing component model: `JFrame`, `JPanel`, components vs containers.
- Use layout managers (Flow/Border/Grid) to structure a UI.
- Build and run a small multi-panel UI skeleton.

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Unit 04: Generics, Lambdas, GUI Swing & Database Connectivity

- Lecture focus: Swing Basics (Components, Containers, Layouts)

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Today: Roadmap

- Swing overview; ‘JFrame‘, ‘JPanel‘, components vs containers

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Today: Roadmap

- Swing overview; ‘JFrame‘, ‘JPanel‘, components vs containers
- Layout managers (Flow/Border/Grid) and UI composition

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Today: Roadmap

- Swing overview; ‘JFrame‘, ‘JPanel‘, components vs containers
- Layout managers (Flow/Border/Grid) and UI composition
- Building a small multi-panel UI skeleton

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Components vs containers

- Components: `JLabel`, `JButton`, `JTextField`, `JTable`, ...

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Components vs containers

- Components: `JLabel`, `JButton`, `JTextField`, `JTable`, ...
- Containers: `JPanel`, `JFrame` content pane

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Components vs containers

- Components: `JLabel`, `JButton`, `JTextField`, `JTable`, ...
- Containers: `JPanel`, `JFrame` content pane
- Compose UI by nesting panels and adding components inside them.

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Layout managers (quick)

- `FlowLayout`: left-to-right (good for button rows)

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Layout managers (quick)

- `FlowLayout`: left-to-right (good for button rows)
- `BorderLayout`: NORTH/SOUTH/EAST/WEST/CENTER (good for screens)

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Layout managers (quick)

- `FlowLayout`: left-to-right (good for button rows)
- `BorderLayout`: NORTH/SOUTH/EAST/WEST/CENTER (good for screens)
- `GridLayout`: rows/cols (good for simple forms)

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A common UI composition pattern

- Root panel: `BorderLayout`

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A common UI composition pattern

- Root panel: `BorderLayout`
- NORTH: title / header

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A common UI composition pattern

- Root panel: `BorderLayout`
- NORTH: title / header
- CENTER: main panel (form/table)

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A common UI composition pattern

- Root panel: `BorderLayout`
- NORTH: title / header
- CENTER: main panel (form/table)
- SOUTH: action buttons

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Example: multi-panel skeleton (1/3)

```
import javax.swing.*;
import java.awt.*;

public class SimpleFormUI {
  public static void main(String[] args) {
    SwingUtilities.invokeLater(() -> {
      JFrame f = new JFrame("CRM Skeleton");
      f.setDefaultCloseOperation(JFrame.EXIT_ON_CLOSE);

      JPanel root = new JPanel(new BorderLayout(10, 10));
```

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Example: multi-panel skeleton (2/3)

```
// continued...
    root.add(new JLabel("Add Resource", SwingConstants.
        CENTER),
            BorderLayout.NORTH);

    JPanel form = new JPanel(new GridLayout(2, 2, 8, 8));
    form.add(new JLabel("Name:"));
    form.add(new JTextField());
    form.add(new JLabel("Type:"));
    form.add(new JTextField());
    root.add(form, BorderLayout.CENTER);
```

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Example: multi-panel skeleton (3/3)

```
// continued...
      JPanel actions = new JPanel(new FlowLayout(FlowLayout
          .RIGHT));
      actions.add(new JButton("Save"));
      actions.add(new JButton("Exit"));
      root.add(actions, BorderLayout.SOUTH);

      f.setContentPane(root);
      f.pack();
      f.setLocationRelativeTo(null);
      f.setVisible(true);
    });
  }
}
```

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Mini Exercises (2–3 mins)

1. Which is a container: `JPanel` or `JButton`?

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Mini Exercises (2–3 mins)

1. Which is a container: `JPanel` or `JButton`?
2. Which layout is best for header + body + footer screens?

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Mini Exercises (2–3 mins)

1. Which is a container: `JPanel` or `JButton`?
2. Which layout is best for header + body + footer screens?
3. Why prefer `pack()` over fixed `setSize()` (often)?

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Key Takeaways

- Swing UI = components inside containers.

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Key Takeaways

- Swing UI = components inside containers.
- Layout managers create robust UIs without hard-coded positions.

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Key Takeaways

- Swing UI = components inside containers.
- Layout managers create robust UIs without hard-coded positions.
- Multi-panel composition scales better than one giant panel.

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Checkpoint Questions

1. What is the difference between a component and a container?

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Checkpoint Questions

1. What is the difference between a component and a container?
2. Name 3 layout managers and one use case for each.

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Checkpoint Questions

1. What is the difference between a component and a container?
2. Name 3 layout managers and one use case for each.
3. Describe one UI screen using the NORTH/CENTER/SOUTH pattern.

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Next Lecture Preview

Next: Swing event handling + JDBC (persistence)

- We will connect buttons to actions and store data in a database.

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